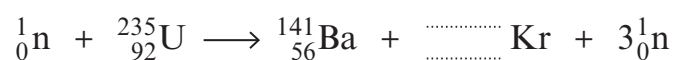


3. Hunterston B is a nuclear power station in Scotland.

(a) The reaction in the reactor is shown in the equation below.



(i) **Complete** the equation. [2]

(ii) Circle the part of the reactor that slows down neutrons so they can be absorbed by uranium nuclei. [1]

moderator fuel rods concrete

(iii) State how many neutrons are produced by the fission of each uranium nucleus. [1]

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(b) Nuclear reactor 3 at the Hunterston B power station was shut down after cracks were found in the graphite bricks. This made it difficult for the control rods to move up and down.

(i) Explain the role of control rods. [2]

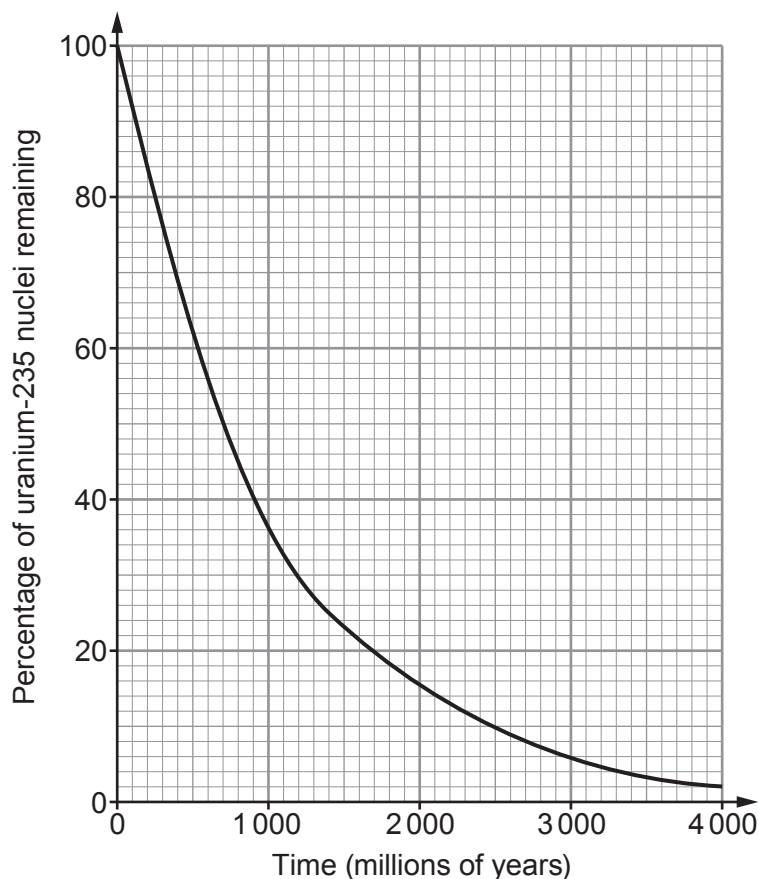
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(ii) An unusual event, such as an earthquake, might move the damaged graphite bricks so that the control rods are unable to move. Explain why this is dangerous. [2]

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- (c) After Hunterston B power station is decommissioned, the uranium-235 in the fuel rods will continue to decay at the rate shown in the graph below.



- (i) Use the graph to answer the following questions.

I. Find the time taken for the uranium-235 to decay from 100% to 50%. [1]

time = million years

II. State the half-life of uranium-235. [1]

half-life = million years

- (ii) The table shows the percentage of uranium-235 nuclei remaining after different half lives. Complete the table. [2]

Number of half-lives	0	1	2	3	4
Percentage of uranium-235 nuclei remaining	100		25		6.25

